Hall Ticket Number:

MR-22

Code: B424CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Software Engineering

(Common to CSE, CSB and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Explain about Software Myths?	1	2	1	1
1. b)	Define CMMI?	1	1	1	2
1. c)	What are the User Requirements?	1	1	2	1
1. d)	Summarize requirement elicitation and analysis?	1	2	2	2
1. e)	Illustrate the Software architecture?	1	2	3	1
1. f)	Construct a Collaboration diagram with your own example?	1	3	3	2
1. g)	Justify the concept of White box testing?	1	5	4	1
1. h)	List few metrics for software quality?	1	4	4	2
1. i)	Determine few Software risks?	1	5	5	1
1. j)	Explain risk projection and how it is used in risk management?	1	4	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain about Changing nature of software and its impact in software engineering.	5	4	1	1
2. b)	Define Process framework and explain in detail its activities.	5	1	1	2
	OR				
2. c)	Why Agile is used and also tell the Agile methodology.	5	1	1	1
2. d)	Explain the Spiral model and Waterfall Model.	5	2	1	2
3. a)	Explain how a software requirement document is structured?	5	2	2	1
3. b)	Distinguish between functional and non-functional requirements?	5	4	2	2
	OR				
3. c)	List the activities of Requirements elicitation and analysis? Explain in detail.	5	4	2	1
3. d)	Explain the Software Requirement Specification (SRS).	5	2	2	2
4. a)	Determine the basic elements of class based model with suitable diagram.	5	5	3	1
4. b)	Distinguish between Coupling and Cohesion? How do they affect software design?	5	4	3	2
	OR				
4. c)	Explain various strategies of design .Which design strategy is most popular and practical?	5	2	3	1
4. d)	Evaluate the various challenges faced during the design process and strategies for overcoming them?	5	5	3	2
5. a)	Explain the art of debugging with sketch?	5	2	4	1
5. b)	Justify the difference between Black box testing and White box testing?	5	5	4	2
	OR				

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5. c)	Explain Software Testing? Discuss role of software testing during software lifecycle?	5	5	4	1
5. d)	What is black box testing? Explain the technique specifying rules and its usage with the help of an example?	5	1	4	2
6. a)	Distinguish between Reactive vs proactive risk strategies?	5	4	5	1
6. b)	What is Risk Refinement? How does it contribute to effective risk management in software projects?	5	1	5	2
	OR				
6. c)	Classify the elements of SQA? Explain in detail?	5	4	5	1
6. d)	Determine the types of software risks? How they affect software quality?	5	5	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome